

1. Student Details

Name	Md Faiz Alam
Course	BCA (Bachelor of Computer Applications)
Semester	5 th Semester
Admission No.	UGO24156735
Project Title	QuizMind AI – Adaptive Quiz Platform for interview & Preparation

2. Work Completed

- Authentication Module: User registration and login system built with email and password. Passwords are securely hashed using SHA-256 before storing in the database.
- Quiz Engine Module: Full quiz screen developed with question display, 4 options, countdown timer (30 seconds per question), correct/wrong answer highlighting, and instant explanation after each answer.
- Question Bank: 894 real interview-style MCQ questions created and seeded into Supabase cloud database across 6 topics and 35 subtopics covering Python, SQL, DSA, Machine Learning, Statistics, and Data Analysis.
- Results Module: After completing a quiz, users see their score, accuracy percentage, correct/wrong count, time taken, and an AI-generated study insight based on performance.
- Analytics Module: Performance dashboard showing total quizzes taken, overall accuracy, topics attempted, weak areas detection (below 60%), bar chart of accuracy by topic, and recent quiz history.
- Database: Cloud PostgreSQL database configured on Supabase with 7 tables: topics, subtopics, questions, users, results, user_answers, topic_scores.
- Backend API: Flask REST API with 3 route groups — auth routes (register, login), quiz routes (fetch questions), result routes (save results, analytics, weak areas, topics, subtopics).
- Frontend: React.js frontend with 5 pages — Login, Dashboard, Quiz, Results, Analytics — connected to Flask backend via REST API calls.

3. System Functionality

The system works in the following steps:

- User registers or logs in on the web applications.
- Dashboard loads all 6 topics and their subtopics fetched from Supabase database.
- User clicks a subtopic — quiz screen loads with questions fetched from Flask API.
- Each question has a 30-second countdown timer and 4 answer options.
- After selecting an answer, correct option is highlighted green, wrong option red, and explanation shown.
- After completing all questions, results are saved to Supabase and results page is shown.
- Results page shows score, accuracy, time taken and AI study suggestion.
- Analytics page fetches user history and shows performance charts and weak areas.

4. Screenshots -> Output

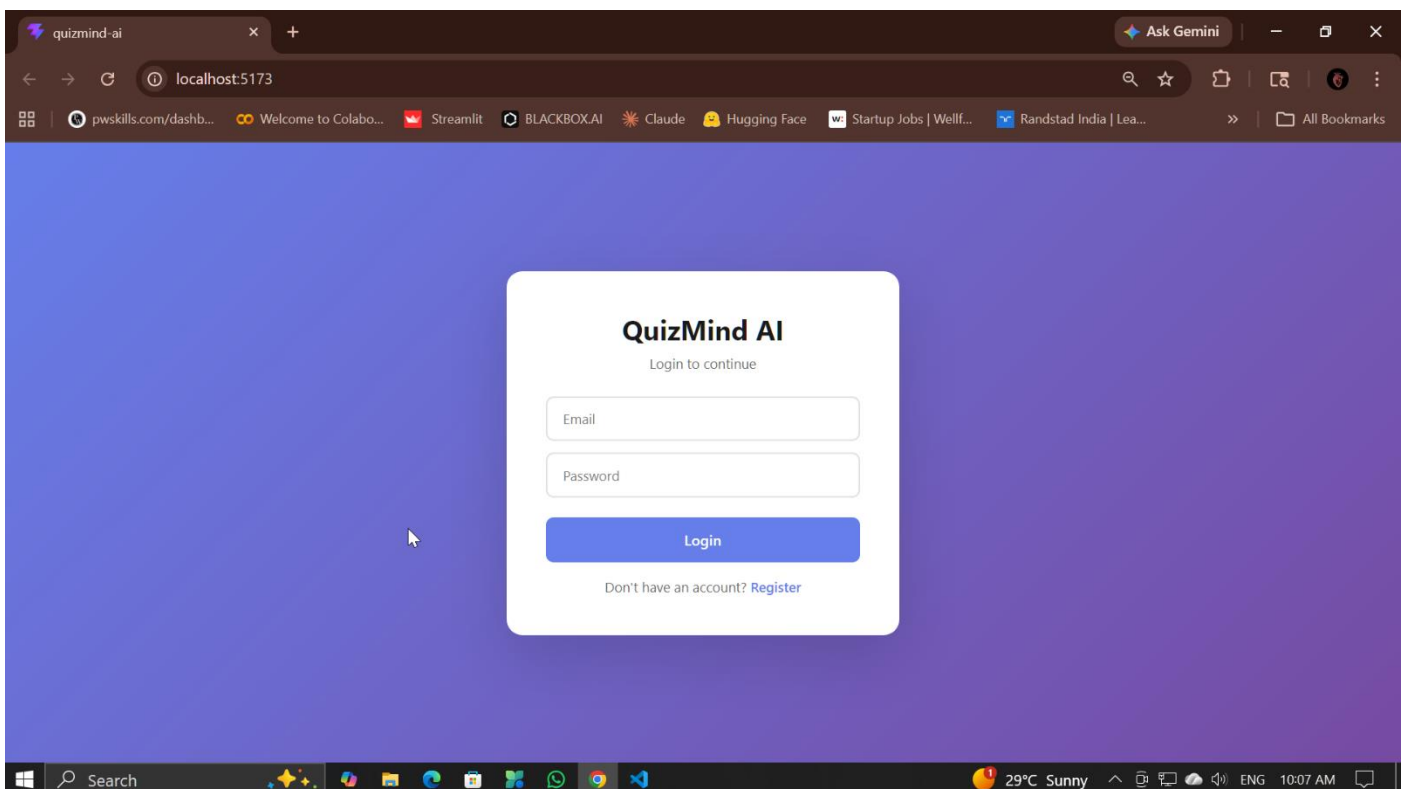


Figure 1: Login Page – User authentication Screen

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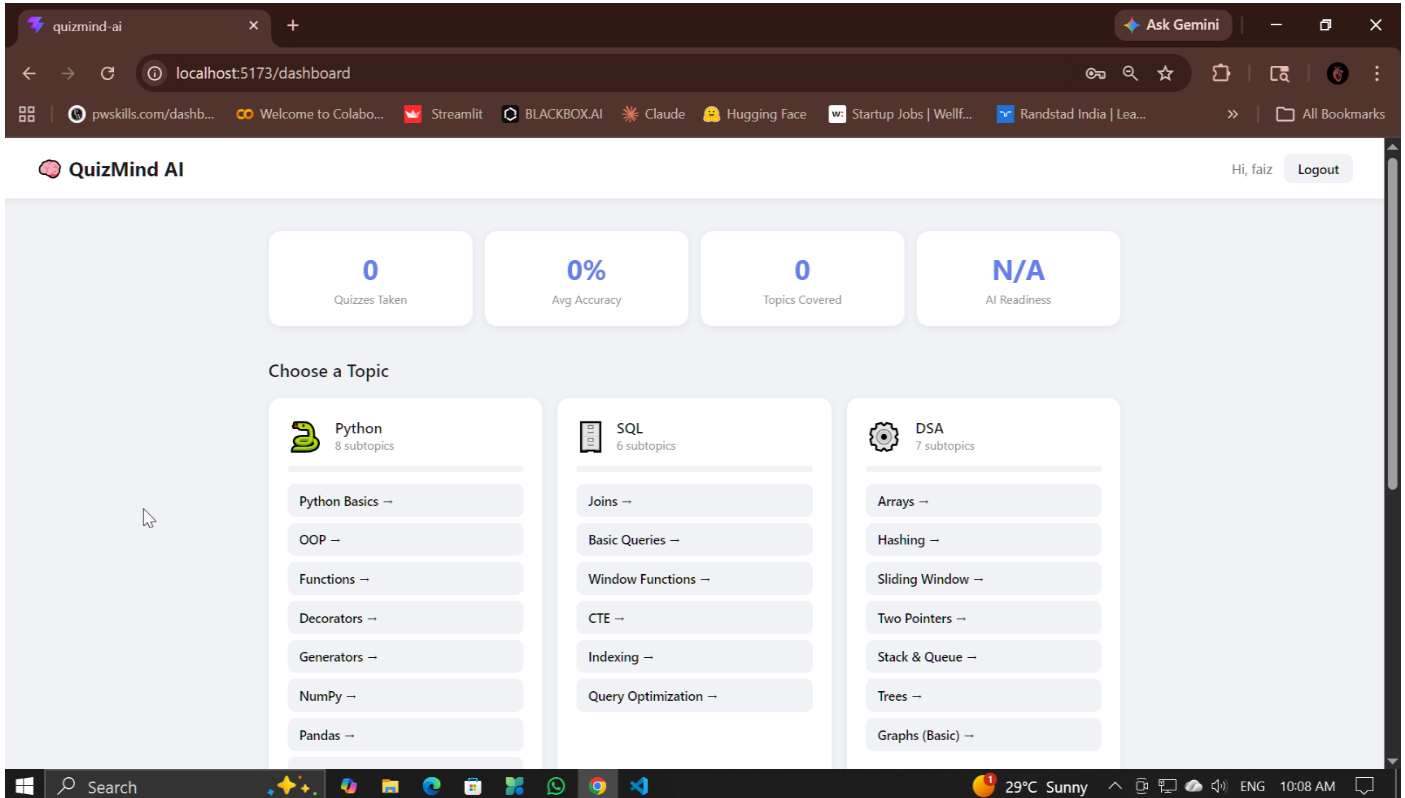


Figure 2: Dashboard – Topic selection with subtopics

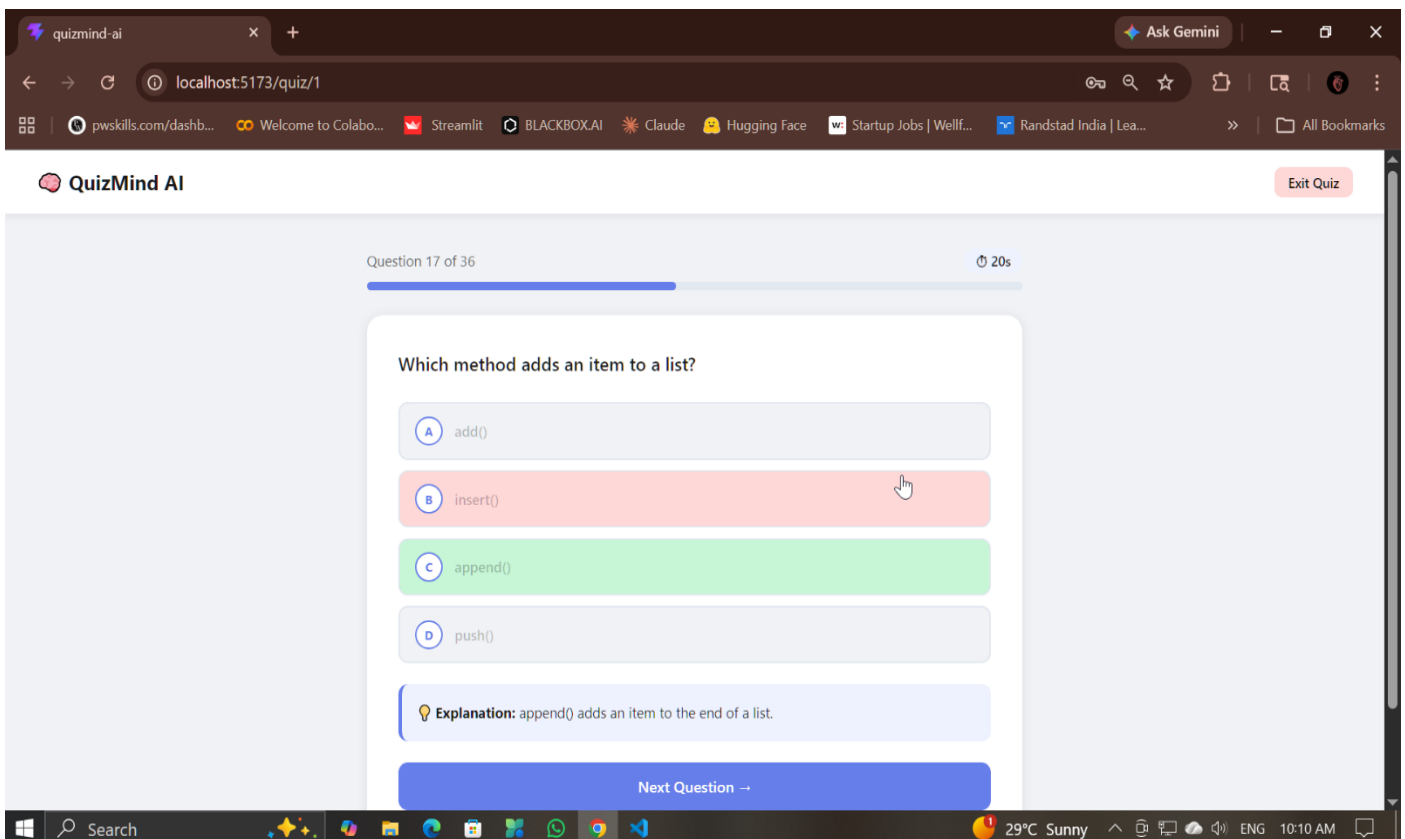


Figure 3: Quiz Screen – Question with timer, options and explanation

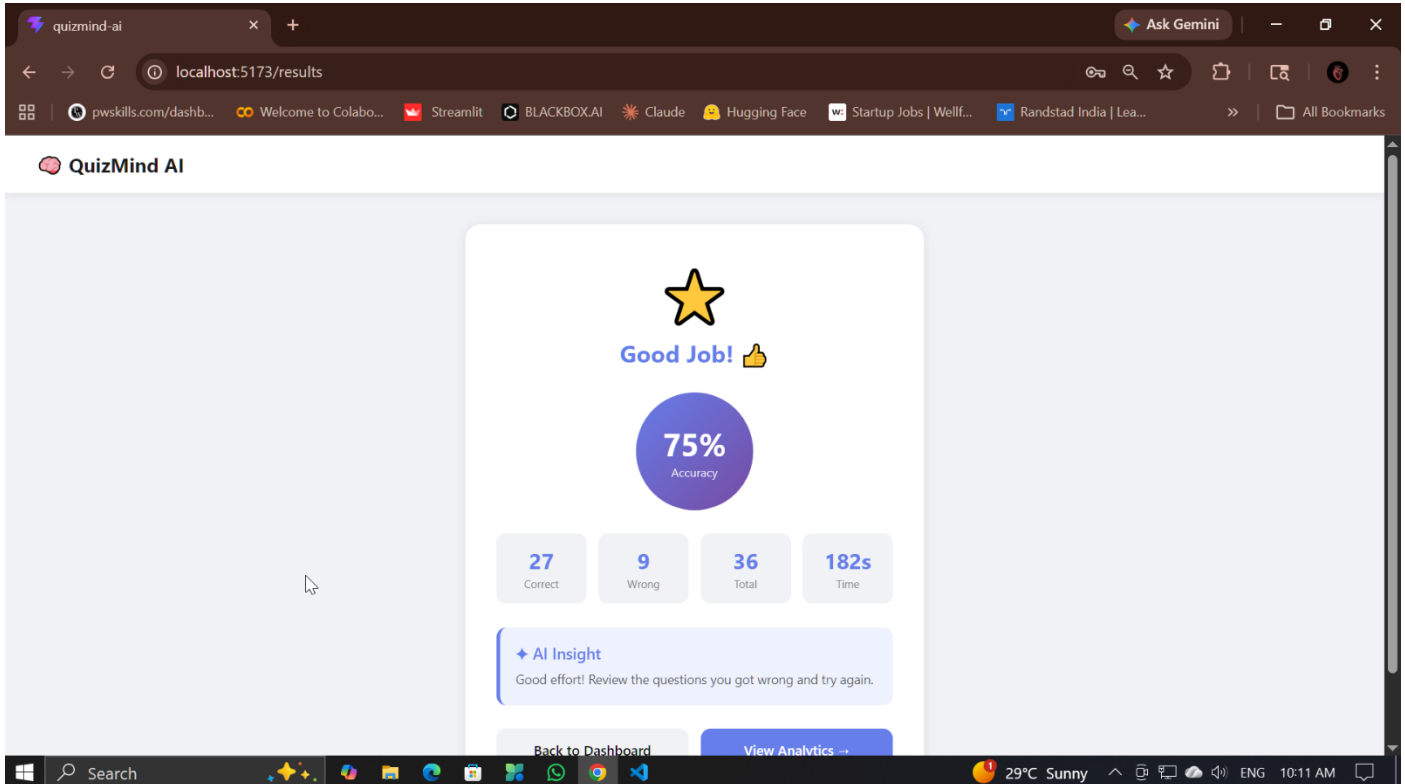


Figure 4: Results Page – Score, accuracy and AI insight

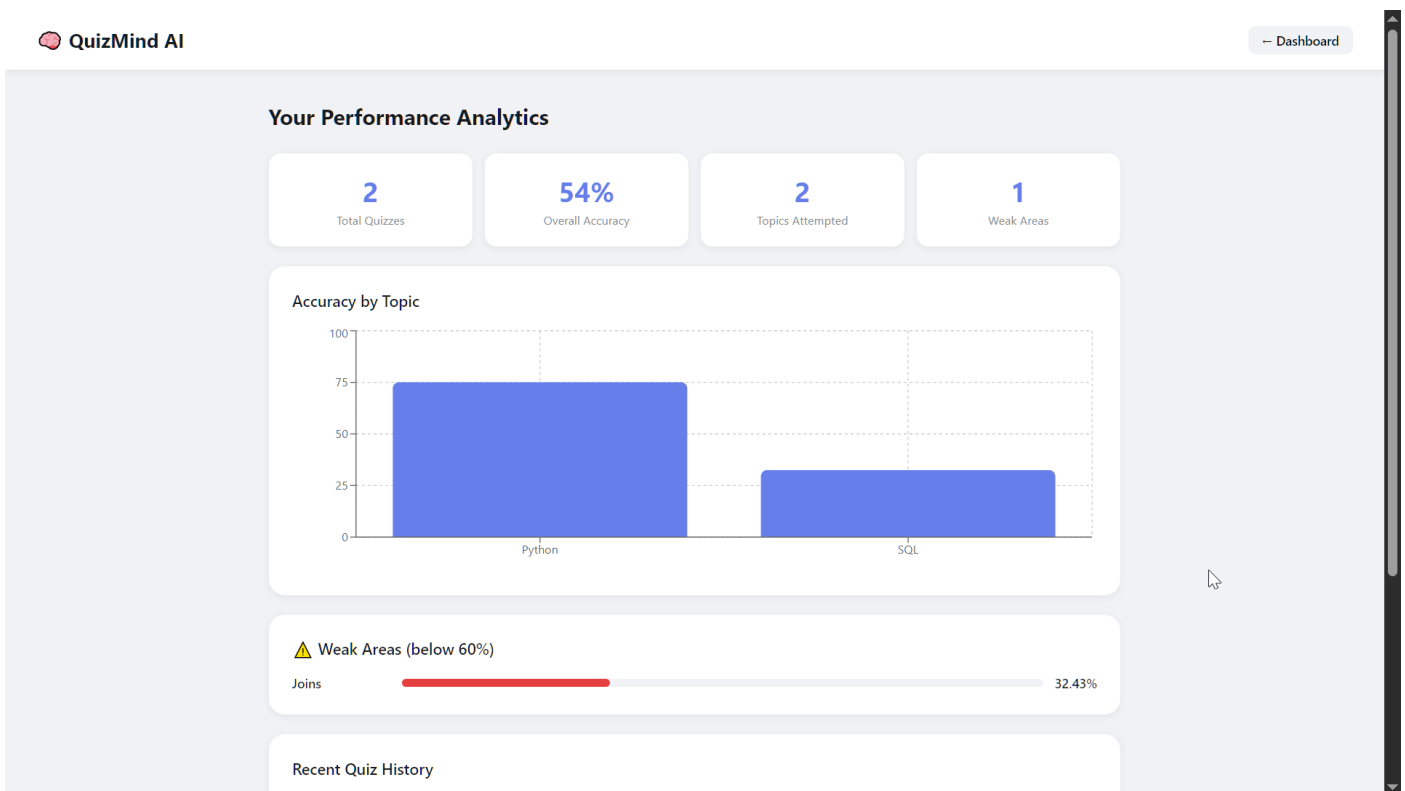


Figure 5: Analytics Page – Performance chart and weak areas

5. Challenges Faced

- Supabase connection timeout — initial connection used wrong port (5432). Fixed by switching to connection pooling port (6543).
- PostgreSQL uses %s placeholder instead of ? used in SQLite — all route queries had to be updated after switching from SQLite to Supabase.
- Data leakage between SQLite and PostgreSQL during migration — resolved by clearing old database and re-seeding all 894 questions fresh.
- React frontend showed white screen due to empty page files — fixed by adding default export to all page components.
- CORS errors when React frontend called Flask backend — resolved by adding Flask-CORS to the backend.

6. Remaining Work

- Deploy Flask backend on Render.com and update frontend API URI
 - Deploy React frontend on Vercel with live backend URL
 - End-to-end testing on deployed live URL
 - Final report preparation and project zip file submission
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